

HTML Fundamentals: Building the Web Structure

Introduction: The Language of the Web

Every website you have ever visited — from social media platforms to online stores to educational portals — is built using HTML. **HTML**, which stands for **HyperText Markup Language**, is the foundation of every web page. It provides the structure and content that browsers display to users.

Think of HTML as the skeleton of a website. Just as a skeleton gives shape and support to a body, HTML gives structure to a web page by organizing text, images, links, and other elements. Without HTML, the web as we know it would not exist.

This reading material examines the fundamentals of HTML, including its history and versions, its role in web development, the structure and syntax of HTML documents, common text formatting elements, and the installation of an Integrated Development Environment (IDE) for writing code. By the end of this lesson, learners will understand how to create basic web pages using HTML.

Part 1: What Is HTML?

Definition and Purpose

HTML is a **markup language** used to create and structure content on web pages. Unlike programming languages that perform calculations or make decisions, HTML describes the structure of a document by using **tags** to mark different parts of the content. A browser reads these tags and displays the content accordingly.

The primary role of HTML is to:

- Organize and categorize content on a website
- Define the structure of a web page (headings, paragraphs, lists, images, links, etc.)

- Provide a foundation that other technologies (like CSS for styling and JavaScript for interactivity) build upon

HTML Versions

HTML has evolved significantly since its creation. Understanding its history helps learners appreciate why certain features exist and why best practices have developed.

Version	Year	Key Features
HTML 2.0	1995	Basic structure, forms, tables
HTML 3.2	1997	Added support for applets, text flow around images, and more formatting options
HTML 4.01	1999	Improved support for stylesheets (CSS), scripting, and accessibility
XHTML 1.0	2000	A reformulation of HTML as an XML language, stricter syntax rules
HTML5	2014	Multimedia support (audio, video), semantic elements, canvas, and improved forms

HTML was originally developed by Tim Berners-Lee while at CERN and popularized by the Mosaic browser. Today, HTML5 is the current standard, supported by all modern browsers.

Part 2: The Structure and Syntax of HTML

Basic HTML Document Structure

Every HTML document follows a standard structure. Below is the basic template for an HTML5 document:

```
html

<!DOCTYPE html>
<html>
  <head>
    <title>Page Title</title>
  </head>
  <body>
    <h1>Main Heading</h1>
    <p>This is a paragraph of text.</p>
  </body>
</html>
```

Explanation of each part:

Element	Purpose
<code><!DOCTYPE html></code>	Declares that this document uses HTML5
<code><html></code>	The root element that contains all other HTML content
<code><head></code>	Contains meta-information about the document (title, styles, scripts)
<code><title></code>	Sets the title shown in the browser tab

Element	Purpose
<code><body></code>	Contains the visible content of the web page
<code><h1></code>	A level 1 heading (largest and most important)
<code><p></code>	A paragraph of text

HTML Syntax: Elements, Tags, and Attributes

Elements and Tags

An **HTML element** consists of:

- An **opening tag**: `<tagname>`
- The **content** inside
- A **closing tag**: `</tagname>`

For example:

```
html
```

```
<p>This is a paragraph.</p>
```

Some elements are **empty elements** and do not have a closing tag, such as `<br>` (line break) and `<img>` (image).

Attributes

Attributes provide additional information about an element. They are written inside the opening tag as name-value pairs:

```
html
```

```
<tagname attribute="value">Content</tagname>
```

Example:

html

```
<a href="https://www.example.com">Visit Example</a>
```

Here, `href` is an attribute that specifies the link's destination.

Common HTML Elements

Element	Tag	Description
Heading	<code><h1></code> to <code><h6></code>	Defines headings (h1 is largest, h6 is smallest)
Paragraph	<code><p></code>	Defines a paragraph of text
Line Break	<code>
</code>	Inserts a single line break (empty element)
Horizontal Rule	<code><hr></code>	Inserts a thematic break (empty element)
Unordered List	<code></code> with <code></code>	Creates a bulleted list
Ordered List	<code></code> with <code></code>	Creates a numbered list
Link	<code><a></code>	Creates a hyperlink
Image	<code></code>	Embeds an image (empty element)

Part 3: Paragraph Formatting Elements

Headings

HTML provides **six levels of headings**, from `<h1>` (most important) to `<h6>` (least important). Headings help structure content and improve readability.

html

```
<h1>Main Title</h1>
<h2>Section Heading</h2>
<h3>Subsection Heading</h3>
<h4>Sub-subsection Heading</h4>
<h5>Level 5 Heading</h5>
<h6>Level 6 Heading</h6>
```

Best Practice: Use headings to create a logical outline of your content. The `<h1>` should be used once per page for the main title.

Paragraphs

The `<p>` tag defines a paragraph. Browsers automatically add space (margin) before and after paragraphs.

html

```
<p>This is the first paragraph. It contains several sentences of text that form a complete thought or idea.</p>
<p>This is the second paragraph. Browsers will add space between paragraphs automatically.</p>
```

Line Breaks and Horizontal Rules

- `<br>` inserts a line break within a paragraph (empty element)
- `<hr>` inserts a horizontal rule (thematic break) to separate content sections

html

```
<p>This is line one.<br>This is line two.</p>
<hr>
<p>This content appears after the horizontal rule.</p>
```

Text Formatting Tags

Tag	Purpose	Example
<code></code>	Bold text (strong importance)	<code>Important</code>
<code></code>	Italic text (emphasis)	<code>Emphasized</code>
<code><u></code>	Underlined text	<code><u>Underlined</u></code>
<code><small></code>	Smaller text	<code><small>Fine print</small></code>
<code><mark></code>	Highlighted text	<code><mark>Highlighted</mark></code>

Example: Formatted Text in HTML

Below is a sample HTML document that demonstrates various paragraph formatting elements. The code shows how different tags affect the appearance of text on a web page.

HTML Code:

```
html

<!DOCTYPE html>
<html>
  <head>
    <title>Formatting Example</title>
  </head>
  <body>
```

```
<h1>Study Tips for Success</h1>
<h2>Effective Learning Strategies</h2>
<p><strong>Remember:</strong> Consistent study habits lead to better retention.</p>
<p><em>Focus on understanding concepts rather than memorizing facts.</em></p>
<p>Use <mark>active recall</mark> to test your knowledge regularly.</p>
<p>Break your study sessions into manageable chunks.<br>Take short breaks between sessions to stay refreshed.</p>
<hr>
<p><small>Disclaimer: Results may vary depending on individual learning styles.</small></p>
<p>This is <u>underlined text</u> for emphasis.</p>
</body>
</html>
```

How It Appears in a Browser:

Study Tips for Success

Effective Learning Strategies

Remember: Consistent study habits lead to better retention.

Focus on understanding concepts rather than memorizing facts.

Use **active recall** to test your knowledge regularly.

Break your study sessions into manageable chunks.
Take short breaks between sessions to stay refreshed.

Disclaimer: Results may vary depending on individual learning styles.

This is underlined text for emphasis.

Explanation of the Tags Used:

Tag	Purpose	Example Output
<code><h1></code>	Main heading (largest)	Study Tips for Success
<code><h2></code>	Section heading	Effective Learning Strategies
<code></code>	Bold text (strong importance)	Remember:
<code></code>	Italic text (emphasis)	<i>Focus on understanding...</i>
<code><mark></code>	Highlighted text	<code><mark></code> active recall <code></mark></code>
<code>
</code>	Line break	Forces text to the next line
<code><hr></code>	Horizontal rule	A line separating content
<code><small></code>	Smaller text	Disclaimer in smaller font
<code><u></code>	Underlined text	<code><u></code> underlined text <code></u></code>

Key Insight: HTML tags control the structure and meaning of content, while the browser determines how to display it visually. Using the appropriate tags makes your content more readable, accessible, and meaningful to both users and search engines.

Part 4: Web Editors and IDE Installation

Choosing a Web Editor

To write HTML code, learners need a **text editor** or an **Integrated Development Environment (IDE)**. Options include:

Editor	Type	Description
Notepad	Basic Text Editor	Simple, built into Windows; minimal features
Notepad++	Advanced Text Editor	Free, syntax highlighting for HTML
Visual Studio Code	IDE	Free, powerful, with extensions for HTML, CSS, and JavaScript
Sublime Text	Text Editor	Lightweight, customizable, paid version available
Atom	Text Editor	Free, open-source, developed by GitHub

Installing Visual Studio Code

Visual Studio Code (VS Code) is a **free, lightweight, and powerful code editor** recommended for web development. It supports HTML, CSS, JavaScript, and many other languages through extensions.

System Requirements:

- Windows 7 or later, macOS 10.12 or later, or Linux
- At least 1 GB of RAM
- 200 MB of available disk space

Installation Steps (Windows):

1. Open a web browser and navigate to: code.visualstudio.com
2. Click the blue **Download for Windows** button
3. Once the download completes, locate the installer file (e.g., `VSCodeUserSetup-x64-*.exe`) in your Downloads folder
4. **Double-click** the installer file to run it
5. Follow the installation wizard instructions:
 - Accept the license agreement
 - Select the installation location (default is recommended)
 - Choose additional tasks (e.g., "Add 'Open with Code' action")
 - Click **Install**
6. Wait for the installation to complete
7. Click **Finish** to launch Visual Studio Code

Installation Steps (macOS):

1. Go to code.visualstudio.com
2. Click **Download for Mac**
3. Open the downloaded `.zip` file
4. Drag `Visual Studio Code.app` to the **Applications** folder

Creating Your First HTML File in VS Code

1. Open Visual Studio Code
2. Click **File** → **New File** (or press `Ctrl+N` on Windows / `Cmd+N` on Mac)
3. Click **File** → **Save As...** (or press `Ctrl+S` / `Cmd+S`)
4. Name your file with the `.html` extension (e.g., `index.html`)
5. Type your HTML code
6. Save your work regularly

Key Terms

Term	Definition
HTML	HyperText Markup Language; the standard language for creating web pages
Tag	A command enclosed in angle brackets (e.g., <code><p></code>) that defines an HTML element
Element	A complete HTML structure consisting of an opening tag, content, and closing tag
Attribute	Additional information added to an HTML tag in name-value pairs (e.g., <code>href="url"</code>)
Web Editor	A software application used to write and edit code
IDE	Integrated Development Environment; a software suite that provides tools for coding, debugging, and testing
DOCTYPE	A declaration that tells the browser which version of HTML the document uses
Heading	An HTML element (<code><h1></code> to <code><h6></code>) used to define section titles
Paragraph	An HTML element (<code><p></code>) used to define a block of text

Review Questions

Write your answers on a separate sheet of paper.

1. What does HTML stand for, and what is its primary purpose in web development?
2. List the six heading tags in HTML. Which one is the largest and most important?
3. Explain the difference between an opening tag and a closing tag. Give an example.
4. What is the purpose of the `<!DOCTYPE html>` declaration?
5. Identify the parts of the following HTML element:

html

```
<a href="https://www.google.com">Visit Google</a>
```

6. Name three different HTML paragraph formatting elements and explain what each does.
7. What is an attribute in HTML? Give one example with its purpose.
8. Why is Visual Studio Code recommended as a code editor for web development?
9. List the basic steps to install Visual Studio Code on a Windows computer.
10. Write the basic structure of an HTML5 document, including `<!DOCTYPE html>`, `<html>`, `<head>`, `<title>`, and `<body>`.

Answer Key

1. HTML stands for **HyperText Markup Language**. Its primary purpose is to create and structure content on web pages.
2. The six heading tags are `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, and `<h6>`. `<h1>` is the largest and most important.

3. An **opening tag** starts an element (e.g., `<p>`), while a **closing tag** ends it (e.g., `</p>`). The closing tag has a forward slash before the tag name.

4. The `<!DOCTYPE html>` declaration tells the browser that the document uses **HTML5**.

5. Parts of the element:

- Opening tag: `<a>`
- Attribute: `href="https://www.google.com"` (specifies the link destination)
- Content: `Visit Google`
- Closing tag: `</a>`

6. Three paragraph formatting elements:

- `<p>` – defines a paragraph of text
- `<br>` – inserts a line break
- `<hr>` – inserts a horizontal rule (thematic break)

7. An **attribute** provides additional information about an element.

Example: `href="url"` in an `<a>` tag specifies the link destination.

8. Visual Studio Code is recommended because it is **free, lightweight, powerful**, supports multiple languages through extensions, and provides features like syntax highlighting, auto-completion, and debugging tools.

9. Basic steps to install VS Code on Windows:

- Go to <https://code.visualstudio.com>[reference:46]
- Click **Download for Windows**
- Run the downloaded installer file
- Follow the installation wizard instructions
- Click **Finish** to launch

10. Basic HTML5 document structure:

`html`

```
<!DOCTYPE html>
<html>
  <head>
    <title>Page Title</title>
  </head>
  <body>
    <h1>Main Heading</h1>
    <p>This is a paragraph.</p>
  </body>
</html>
```